Analysis of ARAS Dataset

1. ARAS Dataset:

1.1 Overview

The **ARAS dataset** differs from many laboratory-based datasets because it was collected in **real residential houses** with residents living their daily lives, rather than actors following a script. This provides a high degree of "ecological validity" but introduces significant noise and complexity.

1.2 System Architecture & Specifications

TABLE II: Properties of the Datasets

	House A	House B
# of PANs	2	1
# of Ambient Sensors	20 of 7 different types	20 of 6 different types
Size of the House	$50m^2$	$90m^2$
House Information	One bedroom, one living room, one kitchen, one bathroom	2 bedrooms, one living room, one kitchen, one bathroom
Residents	2 males both aged 25	Married couple, age average 34
Duration	30 full days	30 full days
# of Activities	27	27

- **Scale:** Data collected from **2 distinct physical houses** (House A and House B).
- **Demographics:**
  - *House A:* 50m^2, occupied by **2 distinct males** (roommates, both aged 25).
  - *House B:* 90m^2, occupied by a **married couple** (Male & Female, average age 34).
- **Duration:** Full **30 days** of continuous recording per house (2 months total data).
- **Sensing Infrastructure:**
  - **Network:** Wireless Ambient Sensor Network (WASN) using the **ZigBee** protocol.
  - **Sensor Count: 20 Binary Sensors** per house.
  - **Sensor Types:** * *Force Sensitive Resistors (FSR) / Pressure Mats:* Located under beds and couches to detect sitting/sleeping.
    - *PIR (Passive Infrared):* Motion detection in rooms.
    - *Contact Sensors (Reed Switches):* On doors, cupboards, and shower cabinets.
    - *Photocells:* Inside drawers/wardrobes to detect opening (light changes).
    - *Temperature Sensors:* On kitchen appliances (e.g., oven).

- *Sonar Distance Sensors:* For presence detection.

1.3 Activities Captured

The dataset includes ground truth labels for **27 distinct activities**.

- **Individual/Simple:** Sleeping, Toileting, Showering, Brushing Teeth.
- **Common Area/Shared:** Watching TV, Studying, Using Internet.
- **Complex/Kitchen:** Preparing Breakfast, Preparing Lunch, Preparing Dinner (Note: The dataset struggles to differentiate the *steps* of cooking, capturing only the general state).
- **Social/Concurrent:** Having a conversation, Having a guest.

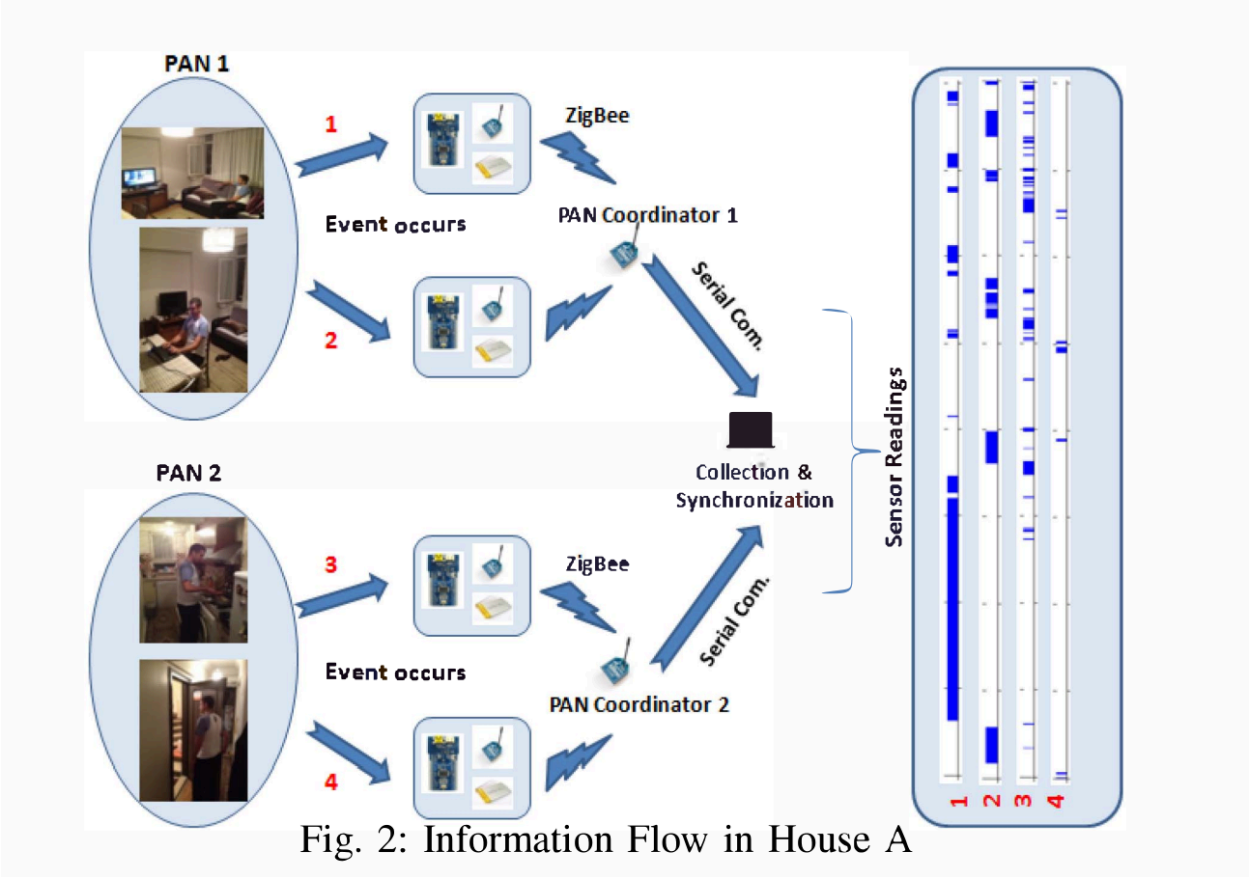


Fig. 2: Information Flow in House A

2. Literature Review: Successes & Baselines

Paper 1: Deep Learning for Multi-Resident Activity Recognition

- **Source:** *Natani et al., "Deep Learning for Multi-Resident Activity Recognition in Ambient Sensing Smart Homes," IEEE GCCE 2019.*
- **Objective:** To benchmark the performance of Recurrent Neural Networks (RNNs) on the ARAS dataset compared to traditional methods.
- **Methodology:**
  - The authors utilized **Long Short-Term Memory (LSTM)** and **Gated Recurrent Units**

(GRU) networks.
    - They treated the sensor data as a time-series sequence, allowing the model to learn temporal dependencies (e.g., "Door open" followed by "Pressure on bed" = Sleeping).
    - They employed **Generative Adversarial Networks (GANs)** to augment the data, creating more synthetic training examples to handle class imbalance.
- **Key Findings (The "Success"):**
    - The models achieved relatively high accuracy for **distinct, sequential activities**.
    - **GRU** outperformed LSTM slightly, achieving approximately **88% accuracy** on House B (the larger house with the married couple).
    - **Takeaway:** This paper proves that for standard, non-overlapping activities, ARAS provides sufficient data for deep learning models to function. If Person A is in the bedroom and Person B is in the kitchen, the system works well.

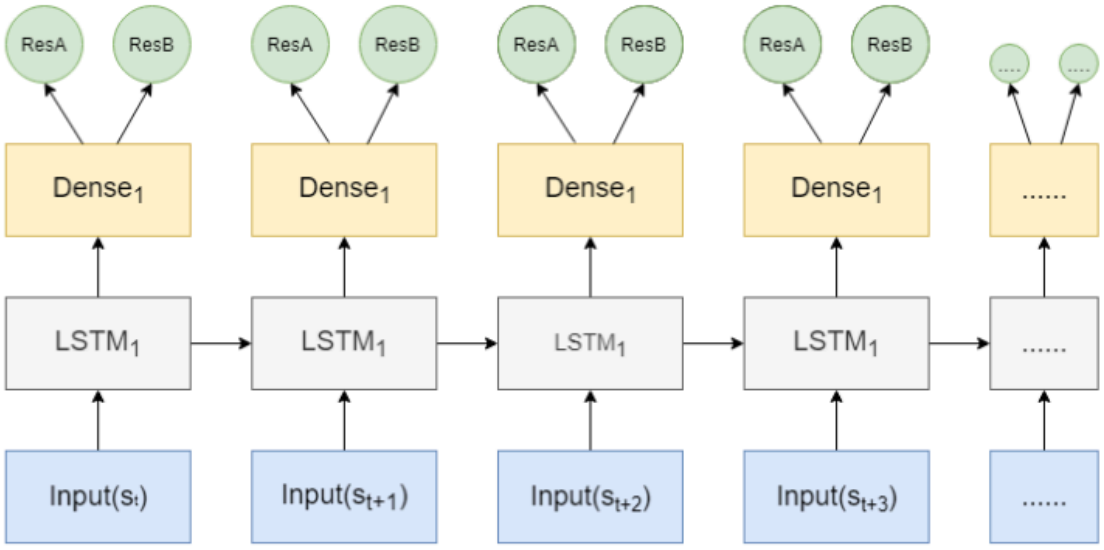


Fig. 1. Model Architecture

Model	Resident A			Resident B		
	10 Days	30 Days	50 Days	10 Days	30 Days	50 Days
LSTM	68.51	71.45	71.61	81.14	86.42	86.55
GRU	70.13	72.56	72.79	83.02	88.14	88.21

3. Literature Review: The Failure Points & Research Gap

While baseline models work for simple scenarios, advanced research highlights critical failures

in the dataset when dealing with complex, multi-user interactions.

Paper 2: *Applying Machine Learning Technologies Based on Historical Activity Features for Multi-Resident Activity Recognition*

- **Source:** *Liang et al., "Applying Machine Learning Technologies Based on Historical Activity Features for Multi-Resident Activity Recognition," Sensors (MDPI), 2021.*
- **Objective:** To improve recognition accuracy by engineering complex "historical features" (activity frequency, duration, period) because raw sensor data was insufficient.
- **The Identified Problem (The Gap):**
  - The authors explicitly state a fundamental flaw in using ambient sensors for multi-resident homes:

    *"Once single-person activity recognition is applied to the multi-person environment, it will fail to identify different residents' activities... [resulting in] an inability to distinguish **which person was involved in what activities**."*
  - **Data Association Ambiguity:** The paper struggled with the fact that ambient sensors are "anonymous." If a motion sensor in the living room triggers, the dataset provides no feature to indicate if it was Resident 1 or Resident 2.
  - **Complex Workarounds:** To fix this, the authors had to build complex probability models based on *historical habits* (e.g., "Resident 1 usually watches TV at 8 PM"). This is not a robust solution for real-time detection of changing behaviors.
- **Corroborating Evidence (MuRAL Paper):**
  - Recent work (MuRAL dataset paper, 2025) further critiques ARAS, noting that it *"does not specify which individual user triggered each event,"* severely limiting its use for tracking individual users during concurrent tasks.

4. Critical Analysis: The Identity Association Gap

Based on the analysis of ARAS and the literature, we have identified the **Identity Association Ambiguity** as the primary bottleneck preventing 100% accuracy in smart home systems.

4.1 The "Same Room" Problem

In ARAS, if Resident A and Resident B are both in the living room:

1. Resident A is reading on the sofa (Pressure Sensor active).
2. Resident B is walking to the window (Motion Sensor active).
3. **System Failure:** The system sees [Pressure=1, Motion=1]. It cannot mathematically determine if Resident A stood up (causing motion) or if Resident B is the one moving. This leads to **inter-class confusion** between "Relaxing" and "Walking."

4.2 Missing Modalities

The failure stems from the lack of **User-Specific Identifiers**.

- **No Wearables:** ARAS has no accelerometer or heart rate data to link movement to a

specific body.
- **No Audio:** ARAS cannot distinguish between the sounds of "chopping food" (User A) and "washing dishes" (User B), leading to confusion in the kitchen.

5. Resolving the Identity Ambiguity Gap:

To construct a **Multi-Modal, Identity-Aware Smart Home Dataset** that resolves the multi-resident ambiguity problem found in ARAS.

We will replicate the ARAS setup (Ambient Sensors) but augment it with two critical layers:

1. **Layer 1: Ambient Sensing (The ARAS Baseline):** Motion, Door, Temperature, and Pressure sensors to capture the *environment's* state.
2. **Layer 2: Wearable/WiFi Sensing (The Solution):**
    - **Wearable BLE Beacons / Smartwatches:** To provide a unique ID (MAC address) for every motion event. This solves the "Who is it?" problem.
    - **WiFi CSI (Channel State Information):** To detect activity intensity without cameras.
3. **Layer 3: Audio (Context):** To differentiate complex parallel activities (e.g., differentiating "Cooking" from "Cleaning" based on sound).

By fusing **Ambient Data** (Location) with **Wearable Data** (Identity), our model will successfully disentangle concurrent activity streams. Where the paper by *Liang et al.* failed to distinguish residents without complex history modeling, our dataset will provide ground-truth identity features, enabling simpler, highly accurate, and real-time multi-resident activity recognition.